

```
In [1]: import numpy as np
```

```
In [2]: a=np.array([[10,15,25],[35,14,12],[36,28,19]])  
print(a,type(a))
```

```
[[10 15 25]  
 [35 14 12]  
 [36 28 19]] <class 'numpy.ndarray'>
```

```
In [3]: a.shape=(9,)  
print(a)
```

```
[10 15 25 35 14 12 36 28 19]
```

```
In [4]: np.sort(a)
```

```
Out[4]: array([10, 12, 14, 15, 19, 25, 28, 35, 36])
```

```
In [5]: np.sort(a)[::-1]
```

```
Out[5]: array([36, 35, 28, 25, 19, 15, 14, 12, 10])
```

```
In [6]: a=np.array([[10,15,25],[35,14,12],[36,28,19]])  
print(a,type(a))
```

```
[[10 15 25]  
 [35 14 12]  
 [36 28 19]] <class 'numpy.ndarray'>
```

```
In [7]: np.sort(a) # By default rowwise sorting
```

```
Out[7]: array([[10, 15, 25],  
               [12, 14, 35],  
               [19, 28, 36]])
```

```
In [8]: a=np.array([[10,15,25],[35,14,12],[36,28,19]])  
print(a,type(a))
```

```
[[10 15 25]  
 [35 14 12]  
 [36 28 19]] <class 'numpy.ndarray'>
```

```
In [9]: np.sort(a,axis=1) # Row wise Sorting
```

```
Out[9]: array([[10, 15, 25],  
               [12, 14, 35],  
               [19, 28, 36]])
```

```
In [10]: a=np.array([[10,15,25],[35,14,12],[36,28,19]])  
print(a,type(a))
```

```
[[10 15 25]  
 [35 14 12]  
 [36 28 19]] <class 'numpy.ndarray'>
```

```
In [11]: np.sort(a,axis=0) # Col wise Sorting
```

```
Out[11]: array([[10, 14, 12],  
               [35, 15, 19],  
               [36, 28, 25]])
```

```
In [12]: a=np.array([[10,15,25],[35,14,12],[36,28,19],[11,75,22]])  
print(a,type(a))
```

```
[[10 15 25]  
 [35 14 12]  
 [36 28 19]  
 [11 75 22]] <class 'numpy.ndarray'>
```

```
In [13]: a.shape=(2,3,2)
```

```
In [14]: print(a)
```

```
[[[10 15]  
  [25 35]  
  [14 12]]  
  
 [[36 28]  
  [19 11]  
  [75 22]]]
```

```
In [15]: np.sort(a)
```

```
Out[15]: array([[10, 15],  
               [25, 35],  
               [12, 14]],  
  
               [[28, 36],  
               [11, 19],  
               [22, 75]])
```

```
In [19]: a=np.array([[10,15,25],[35,14,12],[36,28,19],[11,75,22]])
a.shape=(2,3,2)
print(a,type(a))
```

```
[[[10 15]
  [25 35]
  [14 12]]

 [[36 28]
  [19 11]
  [75 22]]] <class 'numpy.ndarray'>
```

```
In [22]: x=np.sort(a[0],axis=0)
y=np.sort(a[1],axis=0)
```

```
In [23]: print(x)
print(y)
```

```
[[10 12]
 [14 15]
 [25 35]]
[[19 11]
 [36 22]
 [75 28]]
```

```
In [24]: a=np.array([[10,15,25],[35,14,12],[36,28,19],[11,75,22]])
a.shape=(2,3,2)
print(a,type(a))
```

```
[[[10 15]
  [25 35]
  [14 12]]

 [[36 28]
  [19 11]
  [75 22]]] <class 'numpy.ndarray'>
```

```
In [25]: x=np.sort(a[0],axis=1)
y=np.sort(a[1],axis=1)
print(x)
print(y)
```

```
[[10 15]
 [25 35]
 [12 14]]
[[28 36]
 [11 19]
 [22 75]]
```

```
In [ ]:
```

